

iCET CERTIFIED DEVELOPER

Enterprise Application Development

Employee Attendance Management System (EAMS)

Complete Project Documentation & System Manual

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1. Overview of the Project

The **Employee Attendance Management System (EAMS)** streamlines workforce tracking, attendance check-ins/check-outs, leave management, overtime logging, and executive HR analytics. It replaces manual attendance logs with automated status determination (e.g. PRESENT, LATE, EARLY_DEPARTURE) based on configurable shift schedules and grace periods.

Key Objectives

Automated Time & Attendance: Real-time daily check-in and check-out with automated late arrival and early departure flags.

Role-Based Workflows (RBAC):

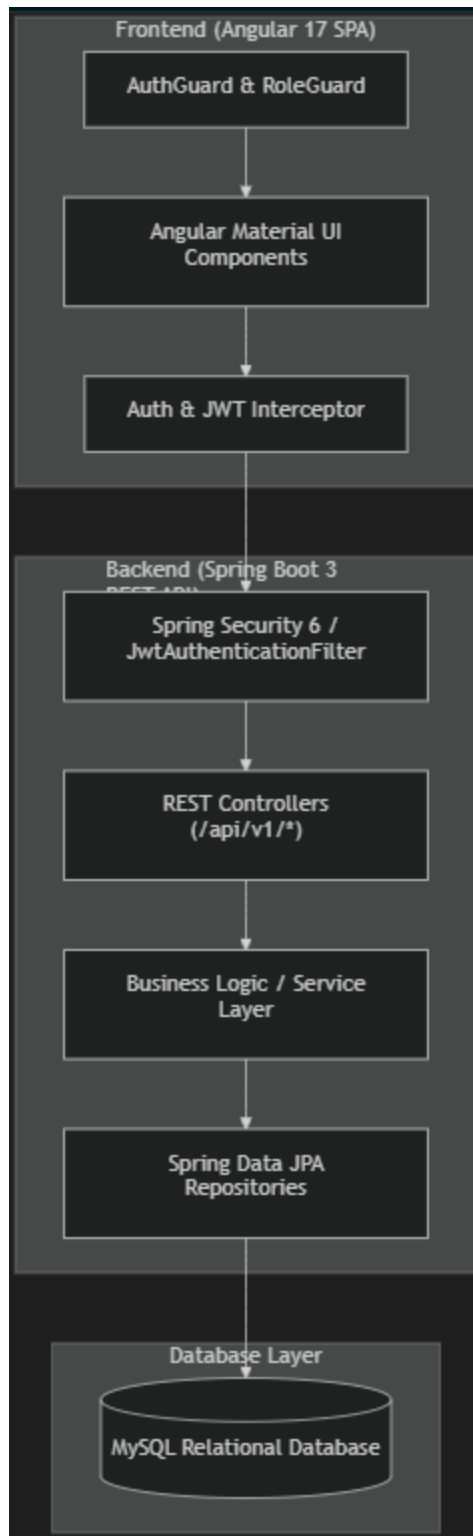
- **Employees:** Perform daily check-in/out, submit leave requests, log overtime, and view personal attendance history.
- **Administrators:** Access executive analytics dashboards, manage employees, configure shifts, set company holidays, and approve/reject leave and overtime submissions.

Stateless JWT Security: Secure REST API endpoints backed by Spring Security 6 and JSON Web Tokens (JWT).

Responsive Enterprise UI: Modern, glassmorphism-inspired Angular Material interface designed for maximum productivity and clarity.

2. System Architecture

EAMS follows a standard **3-Tier Web Application Architecture**:



3. Technologies Used

Frontend Architecture

Technology	Version	Purpose
Angular	17.3.x	Single Page Application Framework (Standalone Components)
Angular Material	17.3.x	Enterprise UI Component Library
RxJS	7.8.x	Reactive Programming & Async Stream Handling
Chart.js / ng2-charts	4.4.x / 5.0.x	Interactive Attendance & Late/Early Trend Analytics
TypeScript	5.4.x	Type-Safe Client Application Logic

Backend Architecture

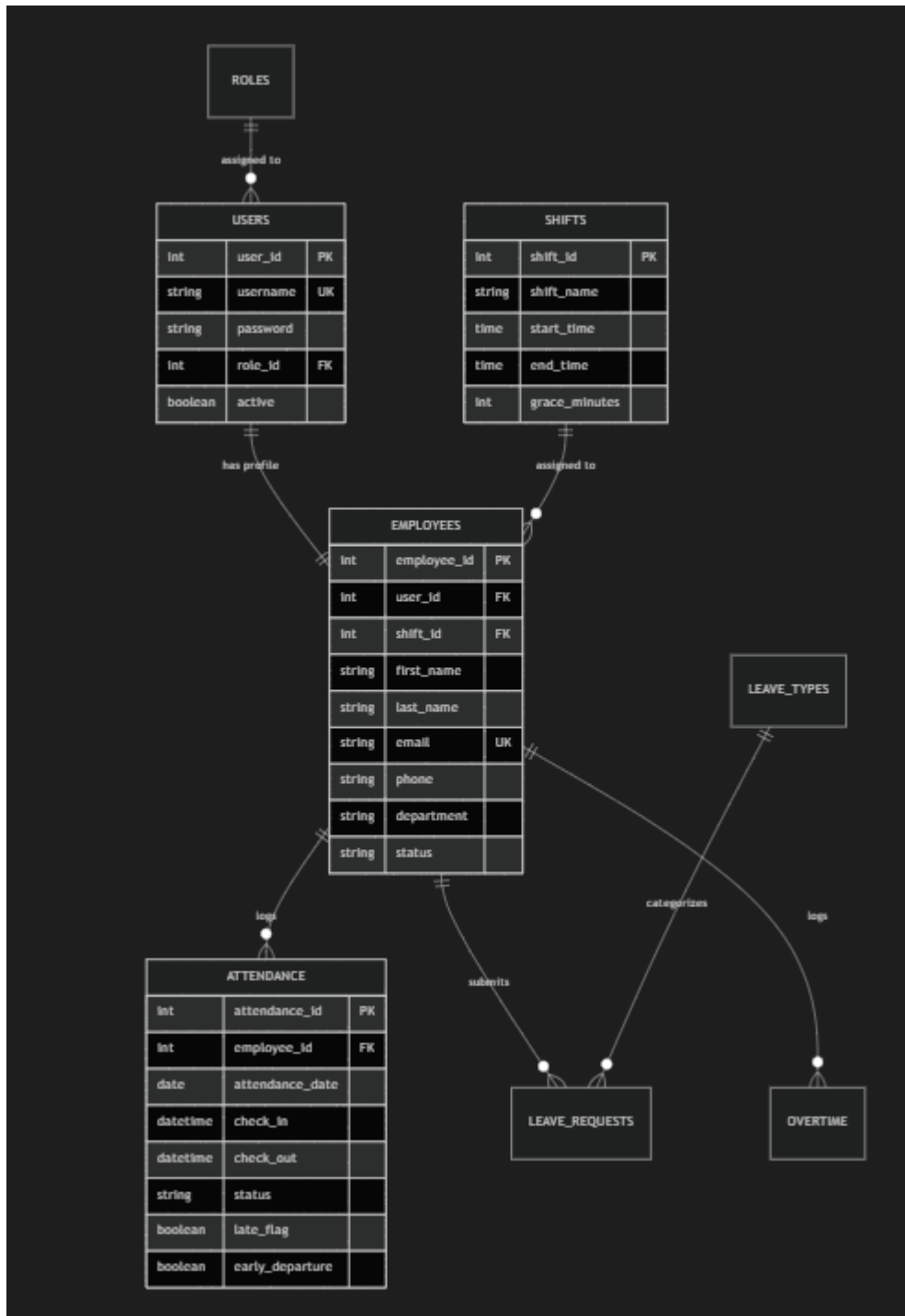
Technology	Version	Purpose
Java JDK	17	Long-Term Support Java Runtime Environment
Spring Boot	3.3.2	Application Framework & Dependency Injection Container
Spring Security	6.3.x	Authentication, Authorization & Security Filter Chains
JWT (io.jsonwebtoken)	0.12.x	HMAC-SHA256 JWT Generation and Parsing
Spring Data JPA / Hibernate	6.5.x	ORM & Relational Database Mapping Layer
Lombok	1.18.x	Boilerplate Code Reduction (Getters, Setters, Constructors)

Database & Infrastructure

Technology	Version	Purpose
MySQL	8.0+	Relational Database Management System
Maven Wrapper (`mvnw`)	3.x	Build Automation & Dependency Management

4. Database Schema & Data Model

The MySQL database `employee_attendance_db` consists of 9 normalized tables:



5. UI Previews & User Interface Design.

Interface Module Previews

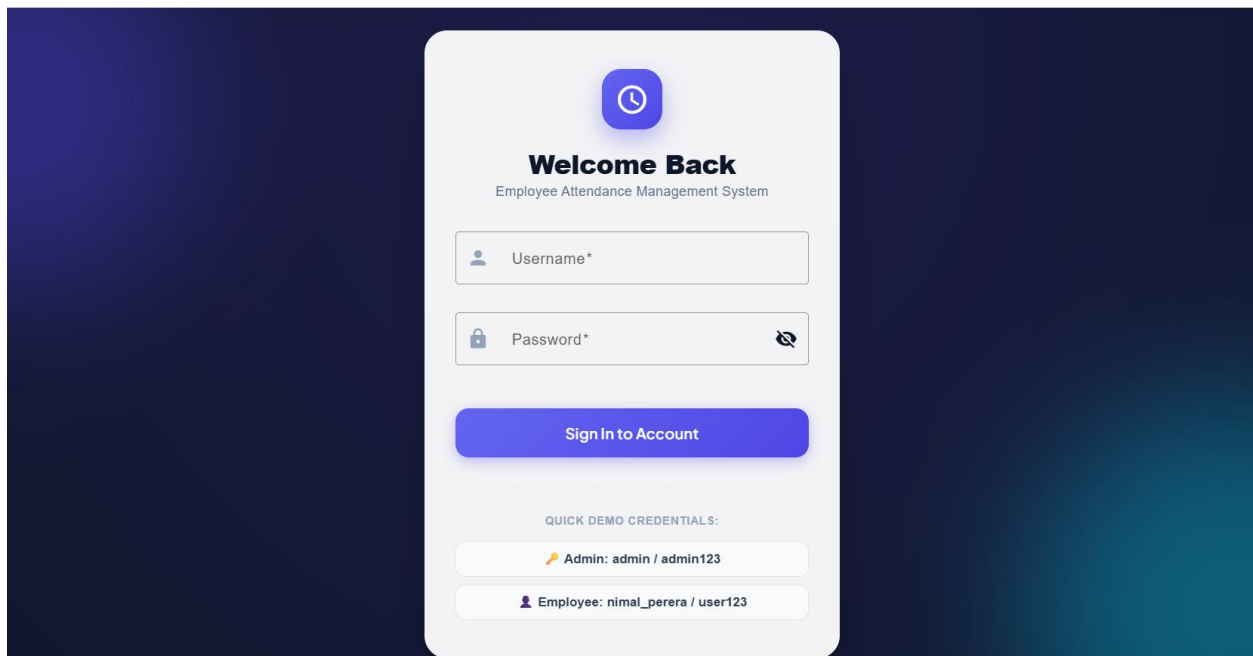
A. Authentication

Features:

- Quick Demo Credentials Chips for rapid testing.
- Password visibility toggle.
- Auto-routing based on role (`ADMIN` -> `dashboard`, `EMPLOYEE` -> `attendance`).

Demo Credentials:

- **Admin:** `admin` / `admin123`
- **Employee:** `nimal\_perera` / `user123`



B. Employee Attendance Portal

Features:

- Live real-time clock displaying current date and local time.
- One-click **Check In** and **Check Out** actions.
- Real-time notification banners for successful check-ins and warnings.

- Attendance history table displaying date, check-in time, check-out time, status badges and flags.

🕒

EAMS Attendance

📅 Attendance

📅 Leave

🕒 Overtime

👤 nimal_perera
EMPLOYEE

➔

🕒

07:47:40 PM

Saturday, July 25, 2026

➔ Check In

➔ Check Out

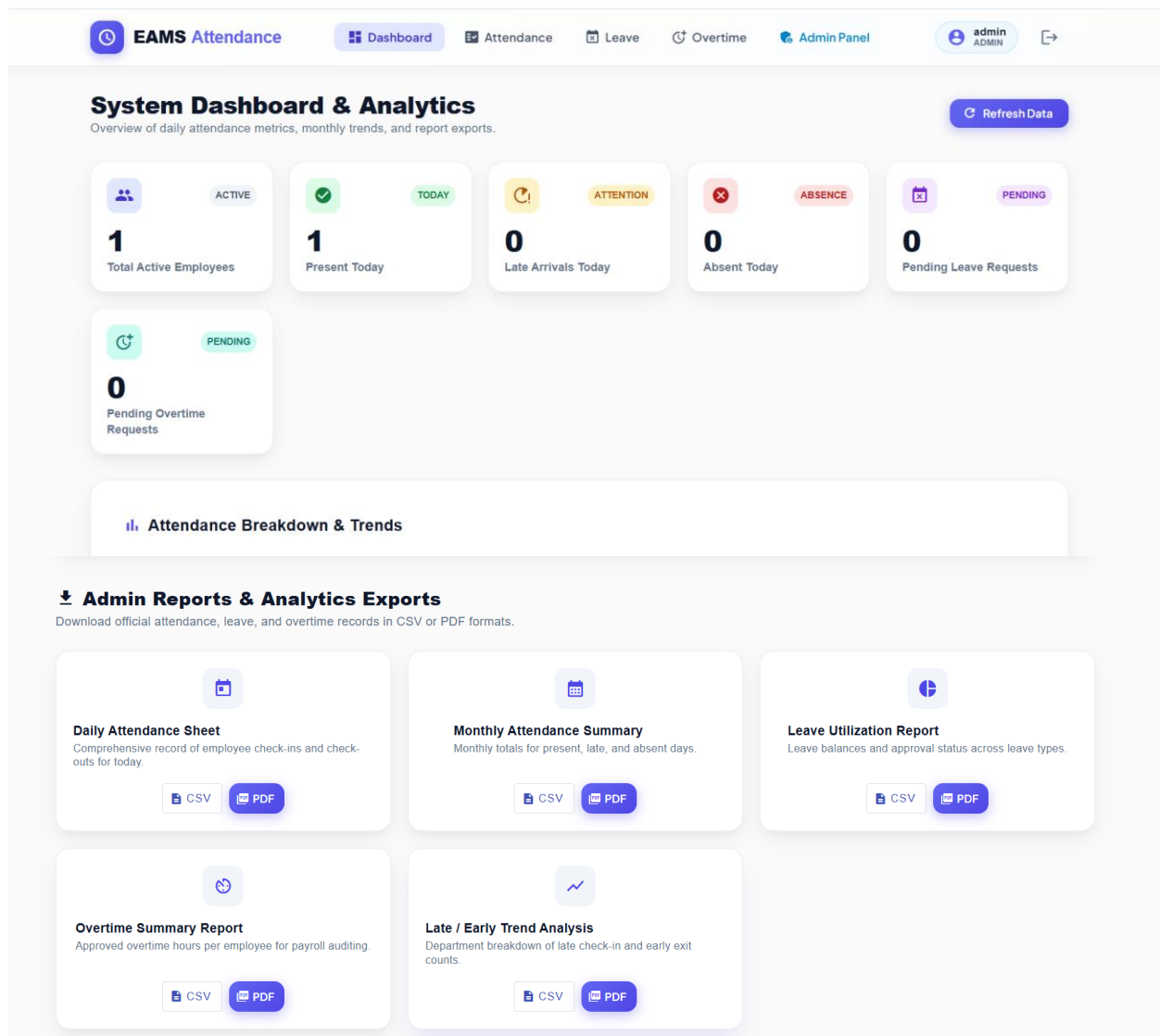
🕒 Personal Attendance History

Date	Check In	Check Out	Status	System Detection
Jul 25, 2026	7:02 PM	7:47 PM	PRESENT	🕒 On Time

C. Executive HR Analytics Dashboard (Admin Only)

Features:

- Metric summary cards for Total Employees, Today's Present Count, Today's Late Count, and Pending Leaves.
- Attendance Trend Chart (Monthly Present vs Late vs Absent).
- Late & Early Arrival Breakdown.
- Department Leave & Overtime distribution graphs.



D. Admin Management Panel

Features:

- Employee Directory: Add new employees, assign shifts, update status (ACTIVE / INACTIVE).
- Shift Configuration: Set start times, end times, and grace periods (eg. 10 mins).
- Company Holidays: Add national/company holidays.



Admin Management Panel

Configure employees, work shifts, and company holiday calendars.

Employees

Work Shifts

Holidays

Register New Employee

Employee ID ...

First Name*

Last Name*

Email Address*

Phone Number

Department*

Save Employee Profile

Employee Directory

ID	Name	Email	Department	Status	Actions
#1	Nimal Perera	nimalperera@gmail.com	Engineering	ACTIVE	Deactivate



Admin Management Panel

Configure employees, work shifts, and company holiday calendars.

Employees

Work Shifts

Holidays

Create Work Shift

Shift Title*

Start Time*

End Time*

--:-- --

--:-- --

Grace Minutes (Late Threshold)*

10

mins

Save Work Shift

Configured Work Shifts



Morning

08:00:00 — 17:00:00 (Grace: 10 mins)



Evening

13:00:00 — 22:00:00 (Grace: 10 mins)



 **EAMS Attendance**

Dashboard

Attendance

Leave

Overtime

Admin Panel

admin
ADMIN

→

Admin Management Panel

Configure employees, work shifts, and company holiday calendars.

Employees

Work Shifts

Holidays

 Add Company Holiday

Holiday Name *

Holiday Date *
mm/dd/yyyy

+ Add Holiday

 Official Holiday List

E. Leave & Overtime Management

Features:

- Leave application form (Casual, Sick, Annual) with date pickers.
- Overtime logging with hours and reasons.
- Admin approval/rejection toggle buttons with real-time status updates.

EAMS Attendance

Attendance

Leave

Overtime

nimal_perera

EMPLOYEE

Request Leave

Leave Type*

Annual Leave

Start Date*

mm/dd/yyyy

End Date*

mm/dd/yyyy

Reason for Leave*

Submit Leave Application

My Leave Requests

Type	Start Date	End Date	Reason	Status	Actions
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EAMS Attendance

Attendance

Leave

Overtime

nimal_perera

EMPLOYEE

Overtime Tracking & Claims

Submit extra hours worked and manage overtime approvals.

Request Overtime

Overtime Date*

mm/dd/yyyy

Number of Hours*

hrs

Reason / Task Accomplished*

Submit OT Request

My Overtime Requests

Date	Hours	Reason	Status	Actions
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6. GitHub Repository Link

<https://github.com/Amasha0503/Employee-Attendance-Management-System.git>